



VEXTRONICS

MECHATRONICS FOR
SUSTAINABLE AGRICULTURE

WORK EXPERIENCE

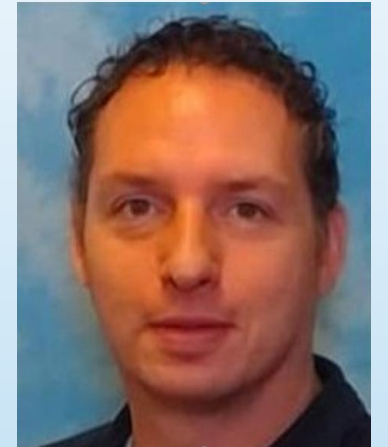


- Education:

- 1997 – 2005 **TU Delft:** Master of Science (MSc.) in Vehicle Mechatronics

- Work Experience:

- 2006 – 2013 **TNO Automotive:** Software design for ABS braking
- 2013 – 2016 **ASML:** Wafer handler Robot controller design
- 2013 – 2015 **TMC:** Electric cooling trailer energy management
- 2016 – 2019 **Cerescon:** Subsurface white asparagus detection
- 2019 – **Vextronics:** System Architect / Consultant Agricultural robotics
- 2021 – **Verfar:** Management of a regenerative farm in Romania



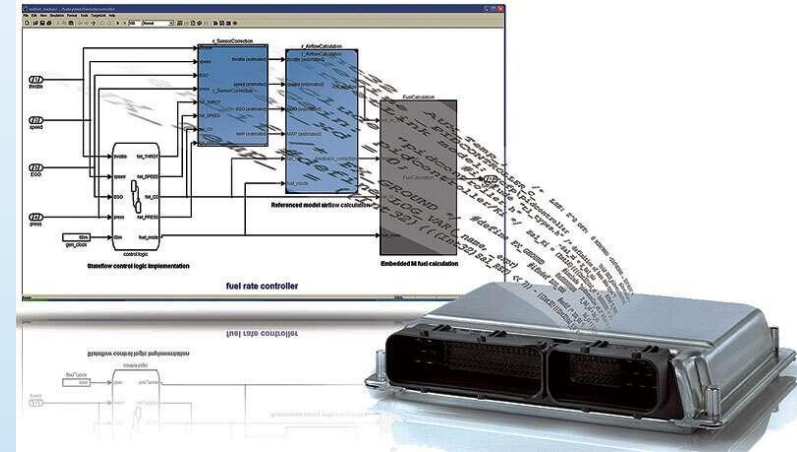
- The dual responsibility Vextronics and Verfar reflects a commitment to merge both the high tech expertise with practical application and knowledge of sustainable farming practices.

MATLAB SIMULINK AUTOMOTIVE EXPERIENCE

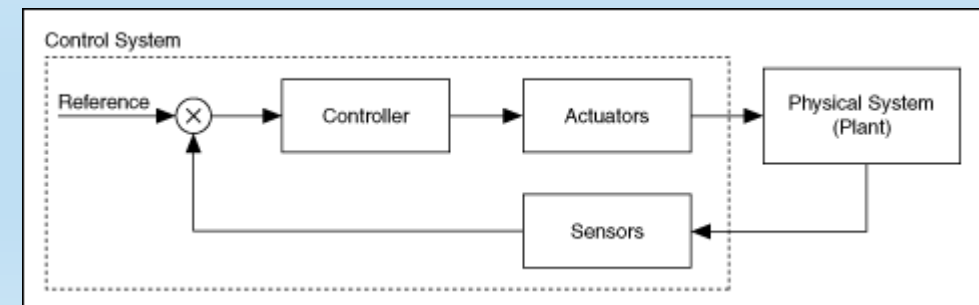


Experience in two “themes”:

- Production code generation
- Controller simulation modelling
- Mostly used toolboxes:
 - Matlab
 - Simulink
 - Stateflow
 - Simulink Coder
 - Control system toolbox
 - Signal Processing toolbox
 - MAB production coding guidelines



“Production code generation”



“Controller simulation modelling”

EMBEDDED CONTROLLER DESIGN EXPERIENCE



- Programming in Structured text & C++:

- Beckhoff IDE TwinCAT 3 <https://www.beckhoff.com/>
- B&R Automation Studio <https://www.br-automation.com/>
- Vectioneer MotorCortex platform <https://vectioneer.com/>



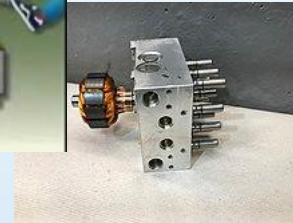
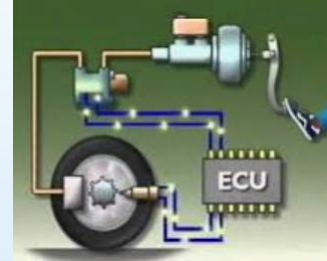
- Rapid control prototyping with Matlab/Simulink:

- dSPACE hardware <https://www.dspace.com/>
- Speedgoat hardware <https://www.speedgoat.com/>



MODEL BASED DESIGN SERVICES

- Software development project examples:
 - **TNO Automotive:** ABS braking system: Hydraulic servo control
 - + Production code generation
 - **ASML:** Wafer handler robot: Servo motor control design
 - + advanced dynamic simulation model
 - **Cerescon:** Asparagus detection system: Advanced sensor fusion algorithm
 - + advanced environmental simulation model
 - **TMC:** Electric cooling trailer: Energy management algorithm
 - + Production code generation



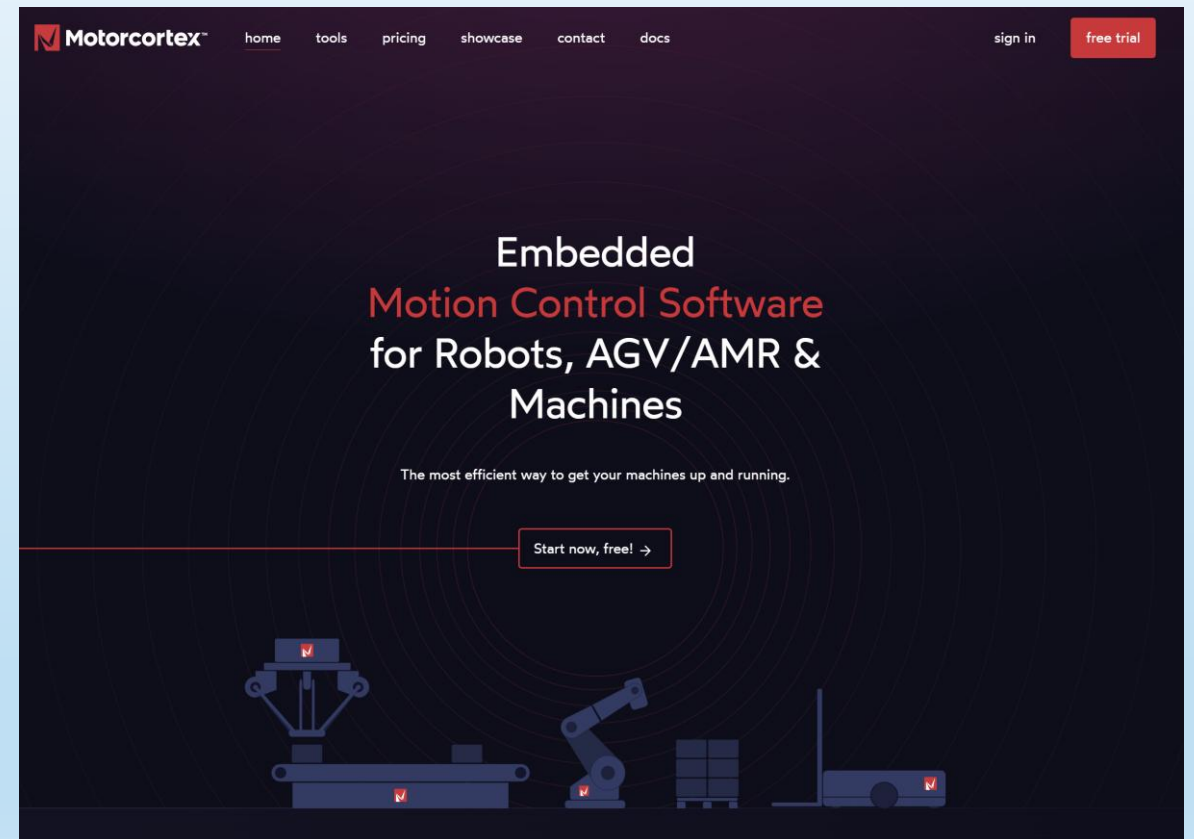
SYSTEMS ENGINEERING SERVICES

- System engineering project examples:
 - **Cerescon:** Design of a sub-surface Asparagus detection system
 - Requirement management
 - Concept generation and decisions
 - **OddBot:** Design of a weed removal robot
 - Requirement management
 - Concept generation and decisions
 - Component selection
 - Prototype testing



VEXTRONICS COLLABORATION WITH MOTORCORTEX

- Web-based automation tools.
- EtherCAT bus-system.
- Ideal for AGV based robots in Food and Agriculture.



<https://motorcortex.io/>



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